

Project Initialization and Planning Phase

Date	29 June 2026
Team ID	N/A
Project Title	OptiCrop Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	Develop an AI-based crop recommendation system that predicts the best crop using soil and weather data to improve agricultural productivity and farmer income.
Scope	The project provides personalized crop recommendations, supports farmers in decision-making, and integrates weather and soil information for accurate predictions.
Problem Statement	
Description	Farmers often select unsuitable crops due to a lack of accurate information about soil fertility and weather conditions, leading to low productivity and financial losses.

Impact	The solution improves crop productivity, reduces farming risks, increases farmer profits, and promotes sustainable agriculture through data-driven decision-making.
Proposed Solution	
Approach	Develop a machine learning model that analyzes soil nutrients (N, P, K), temperature, humidity, rainfall, and pH to recommend the most suitable crop.
Key Features	- AI-based crop recommendation - Weather-based prediction - Soil analysis support

	- User-friendly web interface - Fast and accurate recommendations
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU	Intel Core i5 or above
Memory	RAM	8 GB
Storage	SSD	256 GB SSD
Software		

Frameworks	Python frameworks	Flask
Libraries	ML Libraries	scikit-learn, pandas, numpy, matplotlib
Development Environment	IDE	Jupyter Notebook, VS code
Data		
Data	Source, size, format	Kaggle Crop recommendation Dataset,CSV format